

The Shift Register

The Shift Register is another type of sequential logic circuit that can be used for the storage or the transfer of binary data

This sequential device loads the data present on its inputs and then moves or “shifts” it to its output once every clock cycle, hence the name **Shift Register**.

A *shift register* basically consists of several single bit “D-Type Data Latches”, one for each data bit, either a logic “0” or a “1”, connected together in a serial type daisy-chain arrangement so that the output from one data latch becomes the input of the next latch and so on.

Data bits may be fed in or out of a shift register serially, that is one after the other from either the left or the right direction, or all together at the same time in a parallel configuration.

The number of individual data latches required to make up a single **Shift Register** device is usually determined by the number of bits to be stored with the most common being 8-bits (one byte) wide constructed from eight individual data latches.

Shift Registers are used for data storage or for the movement of data and are therefore commonly used inside calculators or computers to store data such as two binary numbers before they are added together, or to convert the data from either a serial to parallel or parallel to serial format. The individual data latches that make up a single shift register are all driven by a common clock (Clk) signal making them synchronous devices.

Shift register IC’s are generally provided with a *clear* or *reset* connection so that they can be “SET” or “RESET” as required. Generally, shift registers operate in one of four different modes with the basic movement of data through a shift register being:

- **Serial-in to Parallel-out (SIPO)** – the register is loaded with serial data, one bit at a time, with the stored data being available at the output in parallel form.
- **Serial-in to Serial-out (SISO)** – the data is shifted serially “IN” and “OUT” of the register, one bit at a time in either a left or right direction under clock control.
- **Parallel-in to Serial-out (PISO)** – the parallel data is loaded into the register simultaneously and is shifted out of the register serially one bit at a time under clock control.

- **Parallel-in to Parallel-out (PIPO)** – the parallel data is loaded simultaneously into the register, and transferred together to their respective outputs by the same clock pulse.

The effect of data movement from left to right through a shift register can be presented graphically as:

Also, the directional movement of the data through a shift register can be either to the left, (left shifting) to the right, (right shifting) left-in but right-out, (rotation) or both left and right shifting within the same register thereby making it *bidirectional*. In this tutorial it is assumed that all the data shifts to the right, (right shifting).

Serial-in to Parallel-out (SIPO) Shift Register

4-bit Serial-in to Parallel-out Shift Register

The operation is as follows. Lets assume that all the flip-flops (FFA to FFD) have just been RESET (CLEAR input) and that all the outputs Q_A to Q_D are at logic level “0” ie, no parallel data output.

If a logic “1” is connected to the DATA input pin of FFA then on the first clock pulse the output of FFA and therefore the resulting Q_A will be set HIGH to logic “1” with all the other outputs still remaining LOW at logic “0”. Assume now that the DATA input pin of FFA has returned LOW again to logic “0” giving us one data pulse or 0-1-0.

The second clock pulse will change the output of FFA to logic “0” and the output of FFB and Q_B HIGH to logic “1” as its input D has the logic “1” level on it from Q_A . The logic “1” has now moved or been “shifted” one place along the register to the right as it is now at Q_A .

When the third clock pulse arrives this logic “1” value moves to the output of FFC (Q_C) and so on until the arrival of the fifth clock pulse which sets all the outputs Q_A to Q_D back again to logic level “0” because the input to FFA has remained constant at logic level “0”.

The effect of each clock pulse is to shift the data contents of each stage one place to the right, and this is shown in the following table until the complete data value of 0-0-0-1 is stored in the register. This data value can now be read directly from the outputs of Q_A to Q_D .

Then the data has been converted from a serial data input signal to a parallel data output. The truth table and following waveforms show the propagation of the logic “1” through the register from left to right as follows.

Basic Data Movement Through A Shift Register

Clock Pulse No	Q_A	Q_B	Q_C	Q_D
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0

4	0	0	0	1
5	0	0	0	0

Note that after the fourth clock pulse has ended the 4-bits of data (0-0-0-1) are stored in the register and will remain there provided clocking of the register has stopped. In practice the input data to the register may consist of various combinations of logic “1” and “0”. Commonly available SIPO IC’s include the standard 8-bit 74LS164 or the 74LS594.

Serial-in to Serial-out (SISO) Shift Register

This **shift register** is very similar to the SIPO above, except were before the data was read directly in a parallel form from the outputs Q_A to Q_D , this time the data is allowed to flow straight through the register and out of the other end. Since there is only one output, the DATA leaves the shift register one bit at a time in a serial pattern, hence the name **Serial-in to Serial-Out Shift Register** or **SISO**.

The SISO shift register is one of the simplest of the four configurations as it has only three connections, the serial input (SI) which determines what enters the left hand flip-flop, the serial output (SO) which is taken from the output of the right hand flip-flop and the sequencing clock signal (Clk). The logic circuit diagram below shows a generalized serial-in serial-out shift register.

4-bit Serial-in to Serial-out Shift Register

You may think what's the point of a SISO shift register if the output data is exactly the same as the input data. Well this type of **Shift Register** also acts as a temporary storage device or it can act as a time delay device for the data, with the amount of time delay being controlled by the number of stages in the register, 4, 8, 16 etc or by varying the application of the clock pulses. Commonly available IC's include the 74HC595 8-bit Serial-in to Serial-out Shift Register all with 3-state outputs.

Parallel-in to Serial-out (PISO)

The Parallel-in to Serial-out shift register acts in the opposite way to the serial-in to parallel-out one above. The data is loaded into the register in a parallel format in which all the data bits enter their inputs simultaneously, to the parallel input pins P_A to P_D of the register. The data is then read out sequentially in the normal shift-right mode from the register at Q representing the data present at P_A to P_D .

This data is outputted one bit at a time on each clock cycle in a serial format. It is important to note that with this type of data register a clock pulse is not required to parallel load the register as it is already present, but four clock pulses are required to unload the data.

4-bit Parallel-in to Serial-out

As this type of shift register converts parallel data, such as an 8-bit data word into serial format, it can be used to multiplex many different input lines into a single serial DATA stream which can be sent directly to a computer or transmitted over a communications line. Commonly available IC's include the 74HC166 8-bit Parallel-in/Serial-out Shift Registers.

Parallel-in to Parallel-out (PIPO)

The final mode of operation is the Parallel-in to Parallel-out Shift Register. This type of shift register also acts as a temporary storage device or as a time delay device similar to the SISO configuration above. The data is presented in a parallel format to the parallel input pins P_A to P_D and then transferred together directly to their respective output pins Q_A to Q_D by the same clock pulse. Then one clock pulse loads and unloads the register. This arrangement for parallel loading and unloading is shown below.

4-bit Parallel-in to Parallel-out

The PIPO shift register is the simplest of the four configurations as it has only three connections, the parallel input (PI) which determines what enters the flip-flop, the parallel output (PO) and the sequencing clock signal (Clk).

Similar to the Serial-in to Serial-out shift register, this type of register also acts as a temporary storage device or as a time delay device, with the amount of time delay being varied by the frequency of the clock pulses. Also, in this type of register there are no interconnections between the individual flip-flops since no serial shifting of the data is required.

The Universal Shift Register

Today, there are many high speed bi-directional “universal” type **Shift Registers** available such as the TTL 74LS194, 74LS195 or the CMOS 4035 which are available as 4-bit multi-function devices that can be used in either serial-to-serial, left shifting, right shifting, serial-to-parallel, parallel-to-serial, or as a parallel-to-parallel multifunction data register, hence their name “Universal”.

These universal shift registers can perform any combination of parallel and serial input to output operations but require additional inputs to specify desired function and to pre-load and reset the device. A commonly used universal shift register is the TTL 74LS194 as shown below.

4-bit Universal Shift Register 74LS194

Universal shift registers are very useful digital devices. They can be configured to respond to operations that require some form of temporary memory storage or for the delay of information such as the SISO or PIPO configuration modes or transfer data from one point to another in either a serial or parallel format. Universal shift registers are frequently used in arithmetic operations to shift data to the left or right for multiplication or division.

Tutorial Summary

Then to summarise a little about **Shift Registers**

- A simple **Shift Register** can be made using only D-type flip-Flops, one flip-Flop for each data bit.
- The output from each flip-Flop is connected to the D input of the flip-flop at its right.
- Shift registers hold the data in their memory which is moved or “shifted” to their required positions on each clock pulse.
- Each clock pulse shifts the contents of the register one bit position to either the left or the right.
- The data bits can be loaded one bit at a time in a series input (SI) configuration or be loaded simultaneously in a parallel configuration (PI).
- Data may be removed from the register one bit at a time for a series output (SO) or removed all at the same time from a parallel output (PO).
- One application of shift registers is in the conversion of data between serial and parallel, or parallel to serial.
- Shift registers are identified individually as SIPO, SISO, PISO, PIPO, or as a Universal Shift Register with all the functions combined within a single device.

In the next tutorial about **Sequential Logic Circuits**, we will look at what happens when the output of the last flip-flop in a shift register is connected directly back to the input of the first flip-flop producing a closed loop circuit that constantly recirculates the data around the loop. This then produces another type of sequential logic circuit called a *Ring Counter* that are used as decade counters and dividers.